Point & Play: A Mobile AR Campus Tour and Scavenger Hunt for Interactive Campus Exploration

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***Abstract*—Augmented reality (AR) provides new opportunities for creating interactive, location-aware experiences that connect digital content with physical environments. This paper presents Point & Play, a mobile AR campus tour and scavenger hunt application designed to support interactive campus exploration. The application allows users to scan selected campus landmarks and access digital content, including historical information, building descriptions, and interactive scavenger hunt activities. Developed for iOS using Swift, SwiftUI, and ARKit, the system combines image tracking, location-based interaction, and a user-friendly mobile interface to create an engaging tour experience. The project demonstrates the feasibility of using mobile AR to make traditional campus tours more immersive, informative, and interactive. Initial development results show that the system can recognize campus targets, display relevant AR content, and support game-like exploration activities. Future work will focus on expanding the number of supported landmarks, conducting onsite user testing after the prototype is stabilized, integrating richer multimedia content, and adapting the application for additional AR platforms.**

Keywords—Augmented Reality, mobile AR, campus tour, ARKit, interactive exploration

I. Introduction

Campus tours play an important role in helping prospective students, new students, families, and visitors become familiar with university facilities, academic spaces, and campus culture. Traditional campus tours, however, often depend on scheduled guides, printed materials, static maps, or web-based information. These approaches may limit visitors' ability to explore the campus independently, revisit information after the tour, or interact with campus landmarks in a meaningful way. As mobile devices become widely available, augmented reality (AR) creates new opportunities to connect digital content with real-world campus environments [1]. Prior work shows that mobile AR campus tour applications can provide campus visitors with self-guided experiences and support features such as points of interest, location search, navigation, 3D campus buildings, virtual tours, and outdoor games [2].

Augmented reality is especially useful for campus exploration because it overlays digital information onto physical locations through a mobile device camera [1]. Instead of requiring users to read a separate map or website, an AR system presents contextual information directly when users scan or approach a campus landmark. For campus tour and visitor-oriented applications, this connection between the physical environment and digital information helps users learn about unfamiliar spaces in a more intuitive and interactive way [2], [3].

In addition to information delivery, AR supports engagement through game-like interaction. Research on AR and gamification in education suggests that combining immersive visual content with game elements can support learner engagement, motivation, interaction, and participation when the activity is designed around meaningful learning goals [4], [5]. For a campus tour, this means that historical facts, building information, and institutional stories can be presented not only as static descriptions but also as interactive tasks or scavenger hunt activities. This approach makes campus exploration more memorable and appealing, especially for users who are familiar with mobile and game-based applications.

This paper presents Point & Play, a mobile AR campus tour and scavenger hunt application for interactive campus exploration. The application allows users to scan selected campus targets and access relevant digital content, including historical information, building descriptions, and scavenger hunt prompts. The system is developed for iOS using Swift, SwiftUI, and ARKit. By combining mobile AR interaction with a scavenger hunt structure, the project explores how traditional campus tours can be made more interactive and provides a prototype framework for helping users learn about campus landmarks.

The main contributions of this work are application-oriented and system-integration focused. First, the project demonstrates the design and development of an iOS-based AR campus tour application. Second, it integrates campus information with game-like scavenger hunt activities to encourage active exploration. Third, it provides an initial implementation framework that can be expanded with additional landmarks, multimedia content, user testing, and future deployment on more advanced AR platforms.

II. Method

A. System Overview

The Point & Play application is designed as an iOS-based mobile augmented reality system for interactive campus exploration. The goal of the system is to provide a self-guided campus tour experience that allows users to access digital information by scanning selected campus landmarks or visual targets. The target users include prospective students, new students, families, and campus visitors who may want to learn about the campus independently and interactively.

As shown in Fig. 1, the application follows a tab-based user interface workflow. After the launch screen, users enter the main tab view, where they can access the AR tour function and the AR game or scavenger hunt function. In the AR tour workflow,

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users scan a supported campus target, view the corresponding information screen, and then return to the main tab view. This interface structure keeps the navigation simple and allows users to move between informational tour content and interactive game-based exploration.

Fig. 1. User interface workflow of the Point & Play application, showing the launch screen, main tab view, AR tour workflow, information display view, AR game start view, and AR game play view.

B. System Architecture

The application is implemented using Swift. SwiftUI is used for the iOS user interface [6], and ARKit is used for augmented reality functionality [7]. ARKit supports augmented reality experiences by combining device motion tracking, world tracking, scene understanding, and display features for Apple platforms [7]. The system also uses SceneKit to support the visualization layer and AR content rendering [8].

The system architecture is shown in Fig. 2. SwiftUI manages the main user interface and screen navigation, while ARKit handles camera-based AR tracking and visual target recognition. The coordinator serves as a bridge between the AR session and the SwiftUI interface. When ARKit detects a target, the coordinator receives the detection event and passes the corresponding information to the user interface. SceneKit supports the visual output layer by rendering AR-related content or overlays associated with recognized targets.

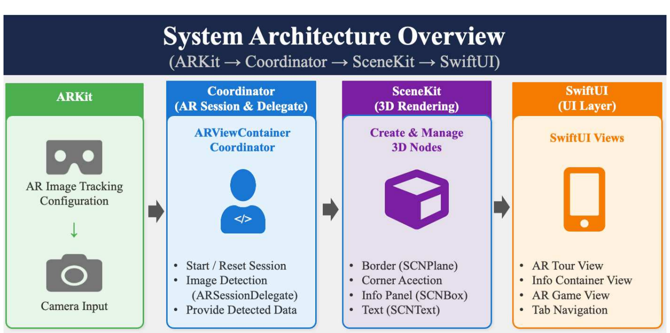

Fig. 2. System architecture of the Point & Play application, illustrating the interaction among ARKit, the coordinator, SceneKit, and SwiftUI.

This architecture separates the application into clear functional layers. The user interface layer manages navigation and information display. The AR processing layer handles image recognition and tracking. The coordination layer connects AR detection events with the application state. The visualization layer presents digital content to the user. This modular design makes the system easier to maintain and expand as more campus locations, AR targets, or interaction features are added.

C. AR Recognition and Content Delivery

The AR recognition process is based on predefined visual targets associated with selected campus locations. Each visual target is linked to campus-related digital content, including building information, landmark descriptions, historical notes, or scavenger hunt clues. This design allows a physical location or visual marker to serve as an entry point to contextual digital information.

The AR image detection pipeline is shown in Fig. 3. The process begins with live camera input from the mobile device. ARKit analyzes the camera view and compares visible images with predefined reference images. When the system detects a supported image, the application maps the detected target to the corresponding campus metadata. The user interface then updates to display the appropriate landmark information or scavenger hunt prompt. ARKit supports this type of image tracking by enabling applications to recognize predefined images in the user's environment and associate them with digital AR content [9].

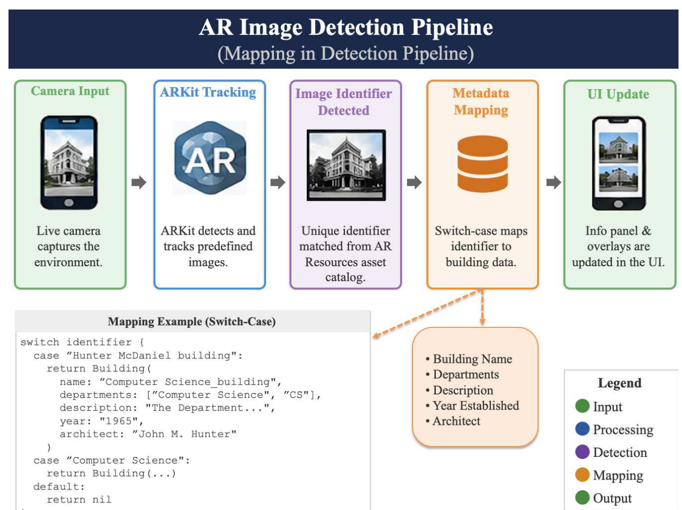

Fig. 3. AR image detection pipeline for mapping recognized campus targets to building metadata and updating the user interface.

This image-based recognition approach provides a direct relationship between a physical campus object and its associated digital information. It also reduces the system's dependence on GPS accuracy, which may be unreliable near buildings, trees, or other dense campus environments. The content delivery process supports both informational and interactive use cases. In the campus tour mode, the recognized target leads to descriptive information about a building or landmark. In the scavenger hunt mode, the recognized target may trigger a clue, prompt, or task that encourages the user to continue exploring other campus locations.

D. *User Interaction and Scavenger Hunt Design*

The user interaction design of Point & Play emphasizes simplicity, accessibility, and engagement. Users interact with the application primarily through the tab-based interface, camera view, and on-screen prompts. After the system recognizes a supported campus target, the application presents the related content in a readable format and allows users to continue exploring other campus locations. This interaction model helps users focus on the campus environment rather than on complex app controls.

In addition to the AR tour function, the application includes a scavenger hunt mode to support active exploration. As shown in Fig. 4, the scavenger hunt interface presents game-like prompts that guide users through campus locations. These prompts can be connected to building features, historical information, or campus stories. Instead of passively reading information, users are encouraged to observe their surroundings, identify landmarks, and complete location-based tasks.

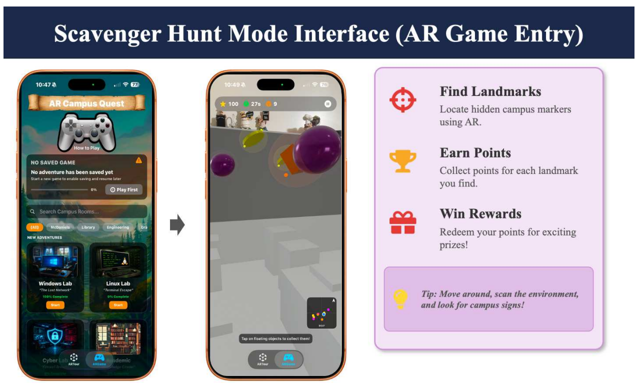

Fig. 4. Scavenger hunt mode interface showing the game entry screen, gameplay view, and interaction tasks such as finding landmarks, earning points, receiving rewards, and exploring campus.

The scavenger hunt design builds on the idea that AR and gamification can support engagement, motivation, interaction, and participation when interactive activities are connected to meaningful learning or exploration goals [5]. In the context of a campus tour, this design transforms campus exploration into a more active and memorable experience. By combining visual recognition, contextual information, and game-like prompts, Point & Play supports both informational learning and interactive discovery.

III. Results

The current implementation of Point & Play serves as a functional prototype and proof-of-concept system. It verifies that the major components can operate together, including the tab-based interface, AR target recognition, metadata-based content retrieval, visualization layer, and scavenger hunt mode. These preliminary results show that the application can connect selected campus targets with relevant digital information and interactive prompts. A formal onsite user study is planned after the prototype is further stabilized.

The application interface provides users with access to both the AR tour and scavenger hunt functions. The tab-based design allows users to move between different modes without navigating through a complex menu structure. This supports the goal of making the application accessible to prospective students, new students, families, and campus visitors. The interface also provides a consistent workflow from launching the application, selecting a mode, scanning a target, viewing related information, and continuing the exploration process.

The AR recognition function successfully connects predefined campus visual targets with corresponding metadata. When the camera identifies a supported image, the system maps the detected identifier to stored campus information and updates the user interface. This result shows that the application's AR recognition pipeline can support location-specific content delivery in the current prototype. It also shows that the system can use image-based recognition as a practical alternative or supplement to GPS-based location triggering [9].

The visualization layer presents digital content after a campus target is recognized. As shown in Fig. 5, the application can display visual output associated with a detected target. This output helps users connect the physical landmark with relevant digital information. In the current implementation, the visual content focuses on demonstrating recognition, metadata mapping, and interface update functions. These results provide a foundation for adding richer AR content, such as 3D models, animations, audio narration, and historical media.

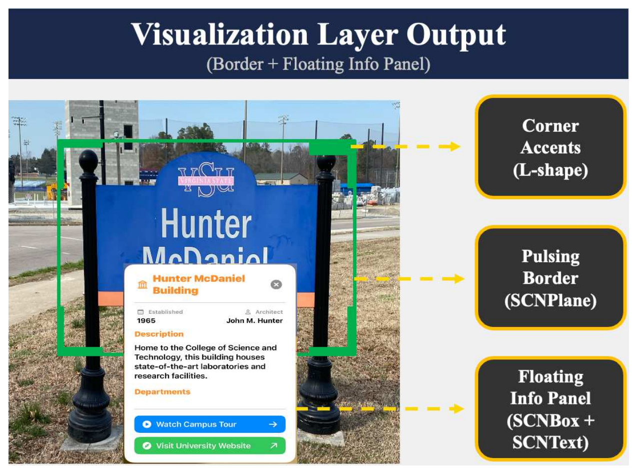

Fig. 5. Visualization layer output showing the AR border, corner accents, and floating information panel displayed after a campus target is recognized.

The scavenger hunt mode demonstrates how the system supports interactive exploration rather than passive information viewing. Through game-like prompts, users are encouraged to identify locations, follow clues, and continue exploring the campus. This mode extends the campus tour experience by giving users a task-oriented reason to move between landmarks. As a result, the application supports both informational learning and active campus engagement.

Overall, the current implementation achieves the major functional goals of the project. It provides an iOS-based AR interface, recognizes predefined campus targets, links targets to campus-related content, and supports a scavenger hunt interaction model. While the current version is an initial prototype, the results demonstrate the feasibility of using mobile AR and image-based recognition to support interactive campus exploration.

IV. Discussion and Conclusion

The Point & Play application demonstrates how mobile augmented reality can support interactive campus exploration by connecting physical campus locations with digital information and game-like activities. Through AR image recognition, the system allows users to scan selected campus targets and access contextual content such as building descriptions, historical information, and scavenger hunt prompts. This design supports a more active and engaging tour experience than traditional static maps, printed materials, or guide-dependent tours.

A key strength of the system is its simple interaction model. Users do not need to learn a complex interface; instead, they interact with the application through a tab-based interface, the mobile camera view, and on-screen prompts. The integration of AR tour content and scavenger hunt activities also allows the application to support both informational and exploratory purposes [4], [5]. This combination is especially useful for prospective students, new students, families, and campus visitors who may want to explore the campus independently.

The current implementation also shows the value of using an image-based AR approach. By connecting predefined visual targets with campus metadata, the system provides a direct relationship between real-world locations and digital content. This approach reduces dependence on GPS-based positioning and supports more precise content triggering when users interact with specific landmarks or visual markers [9]. At the same time, the quality of the experience depends on the availability, visibility, and recognizability of the selected visual targets.

The current work focuses on demonstrating a functional prototype rather than a full user evaluation. This development order is intentional because stable AR recognition, reliable content display, and a usable interaction flow are needed before the system can be evaluated with campus users. Therefore, the results should be interpreted as proof-of-concept evidence of system feasibility, not as measured evidence of improved engagement, learning outcomes, or navigation effectiveness.

In conclusion, Point & Play provides an initial implementation of a mobile AR campus tour and scavenger hunt application. The project demonstrates that ARKit, SwiftUI, and image-based recognition can be combined to create a self-guided and interactive campus tour experience. By integrating campus information with scavenger hunt activities, the application offers a foundation for exploring how campus visits can become more immersive, memorable, and engaging.

V. Future Work

Future work focuses on expanding the application content, improving system robustness, and evaluating the user experience. First, the application can be extended to include more campus landmarks, buildings, and historical locations. A larger set of supported targets will allow users to explore a broader range of campus spaces and make the tour experience more complete.

Second, future development can include richer multimedia content, such as audio narration, images, videos, 3D models, and animated AR overlays. These features can make the tour more informative and visually engaging, especially for locations with important historical or cultural significance.

Third, the scavenger hunt mode can be expanded with additional game mechanics, such as progress tracking, scoring, achievements, timed challenges, and multi-location missions. These features can increase user motivation and provide a more structured exploration experience.

Fourth, future work should include an onsite user study after the prototype is stable enough for deployment. The study can involve prospective students, current students, families, and visitors, and evaluate usability, engagement, perceived usefulness, content clarity, AR recognition reliability, and the scavenger hunt experience. Future evaluation may also compare Point & Play with printed maps, web pages, or guided tour descriptions to assess the impact of mobile AR and game-like interaction on campus exploration.

Finally, the application can be adapted for additional AR platforms and devices. For example, future versions may explore support for Apple Vision Pro or other mixed reality platforms. This direction can provide more immersive campus exploration experiences and extend the system beyond mobile phone-based AR.

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